Alireza **Ghasemian Moghaddam**

M.Sc. Statistics and Data Science, KU Leuven

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Research Interests

Machine learning for clinical decision support, with a focus on multimodal representation learning: fusing structured electronic health record data, medical imaging and free-text clinical notes into a single predictive model. I am particularly interested in how these models behave when a modality is unavailable at prediction time, and in principled statistical evaluation of predictive models in medical settings.

Education

KU Leuven — *Leuven, Belgium*

M.Sc. in Statistics and Data Science — *2025 - 2027 (expected)*

- First-year weighted average: 16.10/20 (80.5%), magna cum laude band
- Master's thesis with Prof. Patrick Onghena (in progress, expected June 2027): Monte Carlo study of the Type I error rate and power of randomization tests that combine evidence across replicated randomized AB (single-case) designs, accounting for the discreteness of their P values

Ferdowsi University of Mashhad — *Mashhad, Iran*

B.Sc. in Computer Science and Engineering — *2019 - 2025*

- GPA: 3.75/4.0 (18.08/20), ranked in the top 10% of a class of 124

Publications

MACD Enhancement through Optimized Weighting and Stability Analysis — *Under review*

A. Ghasemian Moghaddam, T. Bahraini, H. Sadoghi Yazdi — *Submitted 2025*

- Proposes a reweighted formulation of the MACD technical indicator and evaluates its stability across market regimes using a purpose-built backtesting framework.
- Submitted to the *Journal of Economics, Finance and Administrative Science* (Emerald).

Research Experience

Multimodal Fusion of Clinical Data — *Ferdowsi University of Mashhad*

Undergraduate Thesis, supervised by Prof. Hadi Sadoghi Yazdi — *2023 - 2024*

- Built a three-modality fusion network on a cohort of 6,125 intensive care patients from the MIMIC-IV database: clinical time series through a stacked LSTM, chest radiographs through a ResNet-34 pretrained on CheXpert, and free-text discharge and radiology notes through BioWordVec embeddings and a second LSTM, with the modality encoders pretrained separately and then fused end to end.
- 48-hour in-hospital mortality prediction: AUROC 0.883 and AUPRC 0.609, against 0.868 and 0.558 for the strongest single-modality model (chest radiographs) and 0.874 and 0.567 for the published MedFuse two-modality benchmark.
- Phenotype classification over 25 ICD-9 code groups: AUROC 0.772, against 0.755 single-modality and 0.770 for MedFuse.
- Trained with multimodal dropout at $p = 0.3$ so the model tolerates a missing modality at prediction time: with imaging withheld from every sample, AUROC held at 0.882 on mortality and 0.772 on phenotyping.
- Built the full pipeline: cohort extraction and linkage across MIMIC-IV, MIMIC-CXR and MIMIC-IV-Note, per-modality preprocessing, fusion architecture, and evaluation (thesis report).

Automatic Pain Detection from Facial Expressions — *Ferdowsi University of Mashhad*

Research Intern — *2024 - 2025*

- Built a real-time Python pipeline that scores pain frame by frame from video: face detection with RetinaFace, landmark localisation with MobileFaceNet, and facial action unit intensity estimation with gradient-boosted models (Py-Feat), combined into a Prkachin and Solomon Pain Intensity score over AU4, AU6/7, AU9/10 and AU43 and thresholded for a per-frame decision, at five processed frames per second on a live camera feed.
- Ran as the pain-detection module of an ESP32-based knee therapy device: a pain decision stops the device and raises an alert.
- Reviewed the annotated pain corpora available for supervised training (UNBC-McMaster Shoulder Pain Expression Archive, BioVid Heat Pain, BP4D+, Pain Emotion Faces Database) to choose between one-step and two-step designs for patients who cannot self-report.

Expert Backtesting System — *Ferdowsi University of Mashhad*

Undergraduate Researcher — *2020 - 2024*

- Developed a new technical indicator and built a backtesting engine in Python supporting multiple trading strategies and stability analysis across market regimes.
- Work submitted as a journal manuscript (see Publications); implementation released as macdpy.

Technical Skills

Programming	Python, R, SAS, MATLAB, Java, C/C++, SQL
Statistics	Linear and generalized linear models, survival analysis, Bayesian data analysis, multivariate methods, experimental design, sampling theory
ML Frameworks	PyTorch, TensorFlow, scikit-learn, NumPy, pandas
Tools	Git, Linux, Jupyter, Google Colab, LaTeX, WordPress
Languages	Persian (native), English (TOEFL 113/120)

Teaching

Ferdowsi University of Mashhad *Mashhad, Iran*

Head Teaching Assistant, Signals and Systems *Fall 2020, Spring 2021, Fall 2021*

- Supervised every group project in the project-based half of the course, across three semesters.
- Wrote and graded the programming assignments and quizzes, and contributed new course material.

Selected Coursework and Certificates

2025/26	**Data Management**, KU Leuven	*20/20*
2025/26	**Fundamental Concepts of Statistics**, KU Leuven	*19/20*
2025/26	**Statistical Consulting**, KU Leuven	*17/20*
2025/26	**Statistical Software**, KU Leuven	*16/20*
2024	**Neural Networks and Deep Learning (Graduate Course)**, Ferdowsi University of Mashhad	*18.23/20*
2022	**Machine Learning**, Coursera, Stanford Online	*98.03/100*
2022	**Deep Learning Specialization**, Coursera, DeepLearning.AI	*94.05/100*
2022	**Natural Language Processing Specialization**, Coursera, DeepLearning.AI	*100/100*
2022	**Generative Adversarial Networks Specialization**, Coursera, DeepLearning.AI	*100/100*
2022	**TensorFlow: Advanced Techniques Specialization**, Coursera, DeepLearning.AI	*99.33/100*
2022	**TensorFlow: Data and Deployment Specialization**, Coursera, DeepLearning.AI	*98.00/100*

Test Scores

2024	**TOEFL iBT**, Total 113/120 (Reading 30, Listening 30, Speaking 24, Writing 29)	*ETS*
2024	**GRE General**, Total 331/340 (Verbal 163, Quantitative 168), Analytical Writing 4.0/6	*ETS*

Additional Experience

Self-employed *Mashhad, Iran*

Freelance WordPress Developer *2023 - Present*

- Build and maintain WordPress sites for a range of clients, alongside studies.